

CSS Break Properties

break-before, break-after, break-inside: Complete Guide

CSS Guide

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Chapter 1

Introduction

1.1 What are Break Properties?

The CSS break properties (`break-before`, `break-after`, and `break-inside`) control how content breaks across pages, columns, or regions. They define where page breaks, column breaks, or region breaks should occur or be avoided in printed documents and multi-column layouts.

```
/* Basic syntax */
break-before: auto | avoid | all | avoid-page | page;
break-after:  auto | avoid | all | avoid-column | column;
break-inside: auto | avoid | avoid-page | avoid-column;
```

These properties are essential for controlling layout in print media, multi-column designs, and paginated content, ensuring that content breaks logically rather than awkwardly mid-element.

1.2 Why Break Properties Matter

1.2.1 Real-World Use Cases

- **Print Documents:** Control page breaks in invoices, reports, and articles
- **Multi-Column Layouts:** Manage breaks in magazine-style layouts
- **Responsive Design:** Handle content breaks gracefully across devices
- **Long Articles:** Prevent awkward breaks mid-paragraph or within tables
- **Books/Publications:** Professional book layout with chapter breaks
- **Data Tables:** Keep table rows together for readability

1.2.2 Evolution from page-break Properties

These properties replace the older page-break properties:

- `page-break-before` → `break-before`
- `page-break-after` → `break-after`
- `page-break-inside` → `break-inside`

The newer properties are more flexible and support multiple break contexts (page, column, region).

1.3 Browser Support

Break properties have excellent support in modern browsers:

Browser	break-* properties	Status
Chrome/Edge	v50+	Full Support
Firefox	v65+	Full Support
Safari	v13+	Full Support
Opera	v37+	Full Support
IE 11	page-break-* only	Limited

Chapter 2

Core Properties

2.1 break-before Property

2.1.1 What It Does

The **break-before** property specifies whether a page, column, or region break should occur **before** an element.

2.1.2 Values and Meanings

Value	Effect	Context
auto	No break enforced	Default
avoid	Avoid break if possible	All
avoid-page	Avoid page break	Print
avoid-column	Avoid column break	Multi-column
page	Force page break	Print
column	Force column break	Multi-column
all	Force breaks in all contexts	Print + columns

2.1.3 Examples

```
/* New chapter always starts on new page */
.chapter {
    break-before: page;
}

/* Don't break before heading if possible */
h2 {
    break-before: avoid;
}

/* Force column break in magazine layout */
.new-column {
    break-before: column;
}

/* Avoid any break before this element */
```

```
.keep-with-previous {  
    break-before: avoid;  
}
```

2.2 break-after Property

2.2.1 What It Does

The **break-after** property specifies whether a page, column, or region break should occur **after** an element.

2.2.2 Values and Meanings

Value	Effect	Context
auto	No break enforced	Default
avoid	Avoid break if possible	All
avoid-page	Avoid page break	Print
avoid-column	Avoid column break	Multi-column
page	Force page break	Print
column	Force column break	Multi-column
all	Force breaks in all contexts	Print + columns

2.2.3 Examples

```
/* Each section ends with page break */  
.section {  
    break-after: page;  
}  
  
/* No break after paragraphs */  
p {  
    break-after: avoid;  
}  
  
/* Column break after header */  
.column-header {  
    break-after: column;  
}  
  
/* Prevent break after list items */  
li {  
    break-after: avoid;  
}
```

2.3 break-inside Property

2.3.1 What It Does

The **break-inside** property specifies whether a page, column, or region break should occur **inside** an element. This prevents elements from being split across breaks.

2.3.2 Values and Meanings

Value	Effect	Context
auto	Breaks allowed inside	Default
avoid	Avoid break inside if possible	All
avoid-page	Avoid page break inside	Print
avoid-column	Avoid column break inside	Multi-column

2.3.3 Examples

```
/* Keep table on single page */
table {
  break-inside: avoid;
}

/* Keep card together */
.card {
  break-inside: avoid;
}

/* Keep code block together */
pre {
  break-inside: avoid;
}

/* Allow breaks inside (default) */
p {
  break-inside: auto;
}
```

Chapter 3

Practical Examples

3.1 Example 1: Document with Chapter Breaks

```
@media print {  
    /* Force page break before chapters */  
    .chapter {  
        break-before: page;  
        margin-top: 2in;  
    }  
  
    /* Prevent break between heading and content */  
    h1 {  
        break-after: avoid;  
    }  
  
    h2 {  
        break-after: avoid;  
    }  
  
    /* Keep paragraphs together */  
    p {  
        break-inside: avoid;  
    }  
  
    /* Keep tables together */  
    table {  
        break-inside: avoid;  
    }  
  
    /* First chapter doesn't start on new page */  
    .chapter:first-of-type {  
        break-before: auto;  
    }  
}
```

3.2 Example 2: Multi-Column Magazine Layout

```
/* Base layout */
.magazine {
  columns: 3;
  column-gap: 20px;
}

/* Prevent article split across columns */
.article {
  break-inside: avoid;
  margin-bottom: 20px;
}

/* Column break after featured article */
.featured-article {
  break-after: column;
}

/* Keep article heading with content */
.article h2 {
  break-after: avoid;
}

/* Heading never alone at column bottom */
.article h2 {
  break-inside: avoid;
  orphans: 3;
}

/* Pull quote in its own column */
.pull-quote {
  break-before: column;
  break-after: column;
  columns: 1;
}
```

3.3 Example 3: Print-Friendly Invoice

```
@media print {
  /* Each invoice on new page */
  .invoice {
    break-after: page;
    page-break-after: page; /* Fallback */
  }

  /* Last invoice: no break */
  .invoice:last-of-type {
    break-after: auto;
  }

  /* Invoice header stays with content */
  .invoice-header {
```

```
        break-after: avoid;
    }

    /* Keep invoice lines together */
    .invoice-body {
        break-inside: avoid;
    }

    /* Keep table rows together */
    table {
        break-inside: avoid;
    }

    /* Prevent row from splitting */
    tr {
        break-inside: avoid;
    }

    /* Keep total section together */
    .invoice-total {
        break-inside: avoid;
        break-before: avoid;
    }
}
```

3.4 Example 4: Long Article Layout

```
@media print {
    /* Article sections: new page */
    .article-section {
        break-before: page;
    }

    .article-section:first-of-type {
        break-before: auto;
    }

    /* Keep section heading with content */
    .article-section h2 {
        break-after: avoid;
    }

    /* Prevent heading isolation */
    h2, h3 {
        break-after: avoid;
        orphans: 3;
    }

    /* Keep figures together */
    figure {
        break-inside: avoid;
        break-after: avoid;
    }
}
```

```
}

figcaption {
    break-before: avoid;
}

/* Keep lists together */
ul, ol {
    break-inside: avoid;
}

/* List item never alone */
li {
    break-inside: avoid;
}

/* Blockquotes stay intact */
blockquote {
    break-inside: avoid;
}

/* Code blocks stay together */
pre {
    break-inside: avoid;
}
}
```

3.5 Example 5: Responsive Data Table

```
/* Screen layout */
table {
    width: 100%;
    border-collapse: collapse;
}

tr {
    border-bottom: 1px solid #ddd;
}

/* Print layout */
@media print {
    /* Keep table on same page */
    table {
        break-inside: avoid;
    }

    /* Table header repeats on new pages */
    thead {
        display: table-header-group;
    }

    /* Avoid page break in rows */
}
```

```
tr {
    break-inside: avoid;
}

/* Prevent last row split */
tfoot {
    display: table-footer-group;
}

/* Keep row data together */
td {
    break-inside: avoid;
}
}
```

Chapter 4

Advanced Techniques

4.1 Combining Break Properties

4.1.1 Multiple Break Control

```
/* Comprehensive break control */
.keep-together {
    break-before: avoid;
    break-after: avoid;
    break-inside: avoid;
}

/* Section with specific breaks */
.section {
    break-before: page; /* Always new page */
    break-after: avoid; /* Don't break after */
    break-inside: avoid; /* Don't split content */
}

/* Smart heading: avoid if alone */
h2 {
    break-after: avoid;
    orphans: 3;
    widows: 3;
}
```

4.2 With print Properties

```
@media print {
    /* Coordinate with page-break properties */
    .chapter {
        break-before: page;
        page-break-before: always; /* IE fallback */
    }

    table {
        break-inside: avoid;
    }
}
```

```
    page-break-inside: avoid; /* IE fallback */
}

/* Fallback for older browsers */
h2 {
    break-after: avoid;
    page-break-after: avoid;
}
}
```

4.3 Orphans and Widows

4.3.1 Integration with Break Properties

```
@media print {
    /* Prevent orphan/widow lines */
    p {
        orphans: 3; /* Min 3 lines at bottom */
        widows: 3; /* Min 3 lines at top */
    }

    /* Prevent heading alone */
    h2 {
        break-after: avoid;
        orphans: 1;
    }

    /* Keep short content together */
    .short-section {
        break-inside: avoid;
    }
}
```

Chapter 5

Browser Support & Fallbacks

5.1 Modern vs Legacy Properties

Modern	Legacy	Compatibility
break-before	page-break-before	Both work
break-after	page-break-after	Both work
break-inside	page-break-inside	Both work

5.2 Progressive Enhancement

```
@media print {  
    /* Modern browsers */  
    .chapter {  
        break-before: page;  
    }  
  
    /* Fallback for older browsers */  
    .chapter {  
        page-break-before: always;  
    }  
  
    /* Or use both together */  
    table {  
        break-inside: avoid;  
        page-break-inside: avoid;  
    }  
}
```

5.3 Feature Detection

```
/* Check for break-* support */  
@supports (break-inside: avoid) {  
    .card {  
        break-inside: avoid;  
    }  
}
```

```
}  
  
/* Fallback for unsupported */  
@supports not (break-inside: avoid) {  
  .card {  
    page-break-inside: avoid;  
  }  
}
```

Chapter 6

Best Practices

6.1 Key Recommendations

6.1.1 1. Test in Print Preview

Always test in:

- Chrome/Edge: Ctrl+P
- Firefox: Ctrl+P
- Safari: Cmd+P
- Actual printer output

6.1.2 2. Use Specific Values

```
/* Good: Specific control */
@media print {
    table {
        break-inside: avoid;
    }
}

/* Less clear: Using page-break */
@media print {
    table {
        page-break-inside: avoid;
    }
}
```

6.1.3 3. Provide Fallbacks

```
/* Best: Both for compatibility */
.element {
    break-before: page;
    page-break-before: always;
}
```

6.1.4 4. Don't Over-Use avoid

```
/* Bad: Too many avoid restrictions */
p { break-inside: avoid; }
li { break-inside: avoid; }
div { break-inside: avoid; }
span { break-inside: avoid; }

/* Good: Strategic placement only */
table { break-inside: avoid; }
.card { break-inside: avoid; }
pre { break-inside: avoid; }
```

6.1.5 5. Combine with Other Properties

```
@media print {
  /* Comprehensive strategy */
  h2 {
    break-after: avoid;
    orphans: 3;
    widows: 3;
  }

  table {
    break-inside: avoid;
    page-break-inside: avoid;
  }

  .section {
    break-before: page;
  }
}
```

Chapter 7

Common Issues & Solutions

7.1 Issue 1: Breaks Not Occurring

Problem: Page breaks not working as expected.

Causes:

- Not in @media print
- Browser doesn't support property
- Conflicting CSS rules

Solution:

```
@media print {  
    /* Must be in print media */  
    .chapter {  
        break-before: page;  
    }  
  
    /* Include fallback */  
    .chapter {  
        page-break-before: always;  
    }  
  
    /* Check specificity */  
    article.chapter {  
        break-before: page !important;  
    }  
}
```

7.2 Issue 2: Content Orphaned

Problem: Heading or content alone at bottom of page.

Solution:

```
@media print {  
    /* Prevent heading alone */  
    h2 {
```

```
        break-after: avoid;
        orphans: 3;
    }

    /* Keep with next element */
    h2 {
        break-after: avoid;
    }

    h2 + p {
        break-before: avoid;
    }
}
```

7.3 Issue 3: Table Split Awkwardly

Problem: Table breaks mid-row.

Solution:

```
@media print {
    /* Keep table together */
    table {
        break-inside: avoid;
    }

    /* If table too large, keep rows */
    tr {
        break-inside: avoid;
    }

    /* Repeating header */
    thead {
        display: table-header-group;
    }

    tfoot {
        display: table-footer-group;
    }
}
```

7.4 Issue 4: Too Many Page Breaks

Problem: Excessive blank space or page breaks.

Solution:

```
@media print {
    /* Only break before major sections */
    .major-section {
        break-before: page;
    }
}
```

```
/* Avoid break for minor sections */
.minor-section {
    break-before: avoid;
}

/* Use avoid strategically */
p {
    break-inside: avoid;
}
}
```

Chapter 8

Quick Reference

8.1 Property Values Summary

Property	auto	avoid	force
break-before	No break	Avoid break	page/column
break-after	No break	Avoid break	page/column
break-inside	Allow breaks	Avoid breaks	N/A

8.2 Common Patterns

```
/* Chapter/section breaks */
.chapter { break-before: page; }

/* Keep tables together */
table { break-inside: avoid; }

/* Headings with content */
h2 { break-after: avoid; }

/* Cards/boxes */
.card { break-inside: avoid; }

/* Code blocks */
pre { break-inside: avoid; }

/* Lists */
ul, ol { break-inside: avoid; }

/* Figures/captions */
figure { break-inside: avoid; }
```

8.3 Complete Print Stylesheet

```
@media print {
  /* Remove non-essential elements */
}
```

```
nav, .sidebar, .ads { display: none; }

/* Chapter/section breaks */
.chapter, .section {
  break-before: page;
}

/* Heading strategy */
h1, h2, h3 {
  break-after: avoid;
  orphans: 3;
}

/* Keep content together */
table, figure, pre, .card {
  break-inside: avoid;
}

/* Prevent table row split */
tr {
  break-inside: avoid;
}

/* List handling */
ul, ol, li {
  break-inside: avoid;
}
}
```

Thank you for reading!

For more information, visit [MDN Web Docs](#) and [web.dev](#)